Lakeview Secondary School

End-of-Term Examinations, November 2026

Mathematics Paper 2

Grade 10 · Time allowed: 1 hour 30 minutes · Total: 40 marks

Candidate name	
Class	
Candidate number	

Instructions to candidates

- Answer all questions in the spaces provided.
- Show all your working; marks are awarded for method.
- Calculators may be used unless a question says otherwise.
- Give non-exact answers to three significant figures.
- The number of marks for each part is shown in brackets.

Section A

Answer all the questions in this section.

1. Solve the following equations.

(a) $5x - 8 = 3x + 14$ [2]

(b) $x^2 - 7x + 12 = 0$ [3]

(c) $\frac{2}{(x + 1)} = \frac{3}{(2x - 1)}$ [3]

2. The table below shows the heights of 40 students in a Grade 10 class.

Height h (cm)	$145 \le h < 150$	$150 \le h < 155$	$155 \le h < 160$	$160 \le h < 165$	$165 \le h < 170$	$170 \le h < 175$
Frequency	3	7	11	10	6	3

(a) Write down the modal class. [1]

(b) Calculate an estimate of the mean height. [4]

(c) Find the probability that a student chosen at random is at least 165 cm tall. [2]

3. A water tank in the shape of a cylinder has a radius of 1.2 m and a height of 2.5 m.

(a) Calculate the volume of the tank in cubic metres. [3]

(b) Water flows out of the full tank at 15 litres per minute. How long, in hours and minutes, does it take to empty? [4]

Section B

Answer all the questions in this section. Diagrams are not drawn to scale.

4. A straight line passes through the points A(−2, 5) and B(4, −7).

(a) Find the gradient of AB. [2]

(b) Find the equation of the line AB in the form $y = mx + c$. [2]

(c) Find the coordinates of the midpoint of AB. [2]

5. A shop increases the price of a jacket by 20% and later reduces the new price by 15%.

(a) The original price was \$80. Find the final price. [3]

(b) Find the overall percentage change in price. [2]

6. In triangle PQR, PQ = 8 cm, QR = 11 cm and angle PQR = 52°.

(a) Calculate the length of PR. [4]

(b) Calculate the area of triangle PQR. [3]

End of examination. Check your answers before you hand in this paper.